•**Feedback System:** A feedback module was implemented to collect and manage faculty performance ratings.

ADHIYAMAAN COLLEGE OF ENGINEERING (AN AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CAMPUS MATE – A COLLEGE AI

ABSTRACT

Campus Mate – A College AI is a smart web-based application developed using Django to improve academic and administrative processes in colleges. It integrates modern web technologies with an AI-powered chatbot to provide quick and efficient support to students, staff, and administration, reducing manual work and enhancing communication.

A key feature is the AI chatbot, which acts as a virtual assistant by providing instant responses to queries related to syllabus, academic procedures, and general college information. This enables students to access information quickly without relying on staff.

The system also includes a Student Navigation Module that helps users easily explore sections like syllabus, assignments, results, and campus directions through a user-friendly interface.

Additionally, the No Due Verification System digitizes the clearance process, allowing staff to verify records and the office to approve them. Students can track their status in real time, ensuring transparency and reducing delays caused by paperwork.

OBJECTIVE

•Automate Query Assistance: Provide instant responses to student location-related questions within the campus.

•Integrate NLP Technology: Understand and process user input using Natural Language Processing techniques.

•Enable Navigation Support: Guide new students and visitors by supplying relevant location information through chat interaction.

•Design User Interface: Develop an interactive and user-friendly web dashboard.

ARCHITECTURAL DESIGN

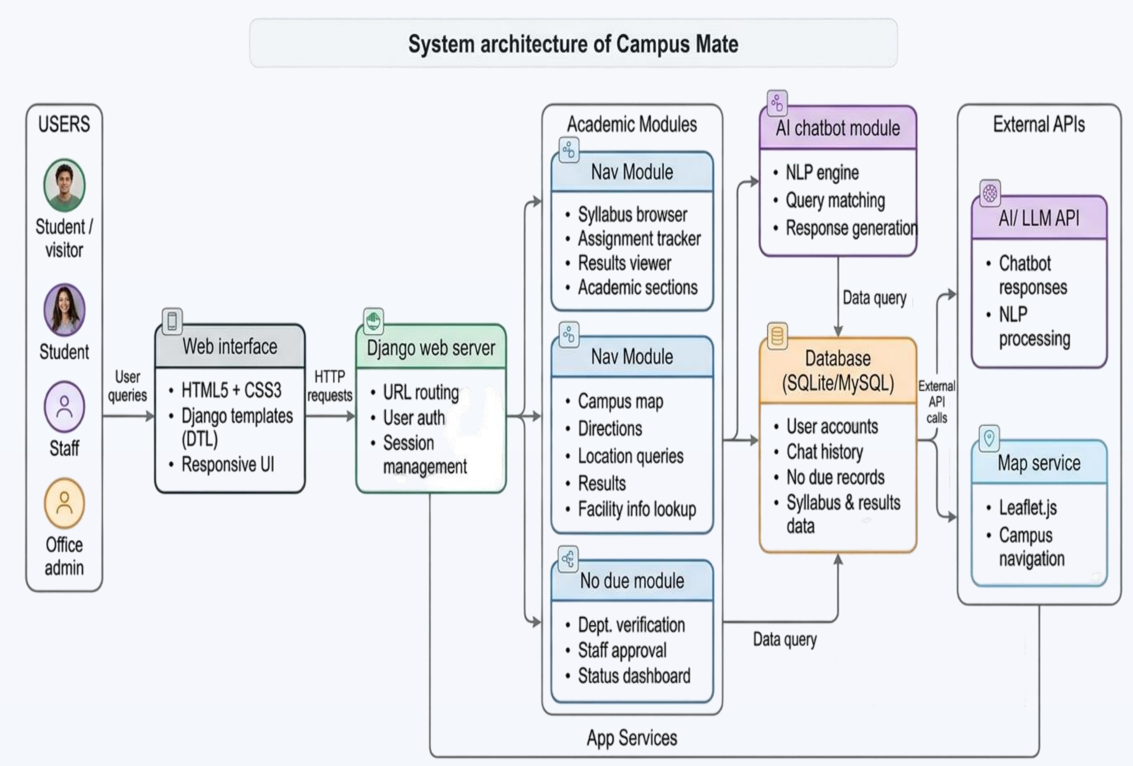


Fig. 1: Architecture of Campus Mate – A College AI

METHODOLOGY

1. User Login and Authentication

Users access the system through a web interface and enter their credentials. The system validates login details and grants secure access to the chatbot dashboard.

2. Query Input

After login, users type location-related questions into the chat interface. These queries serve as input for the chatbot processing system.

3. Request Transmission

The user query is sent from the frontend interface to the backend server for processing through HTTP request handling.

4. Text Processing and NLP Handling

The system processes the query using Natural Language Processing techniques.

5. Response Generation

Based on the matched query, the chatbot generates an appropriate answer containing campus location details or navigation guidance.

6. Navigation Assistance

The system provides text-based directions and guidance to help new students and visitors locate facilities within the campus environment.

7. Chat History Handling

User interactions are maintained within the session, allowing conversation continuity and improved user experience.

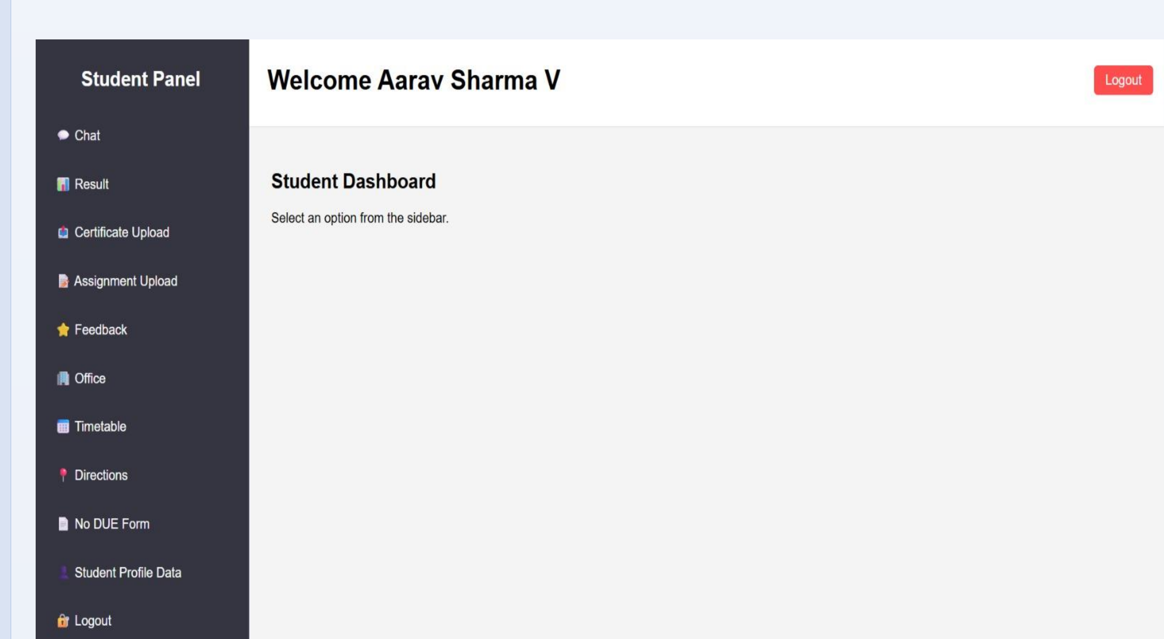
8. Additional Modules Development

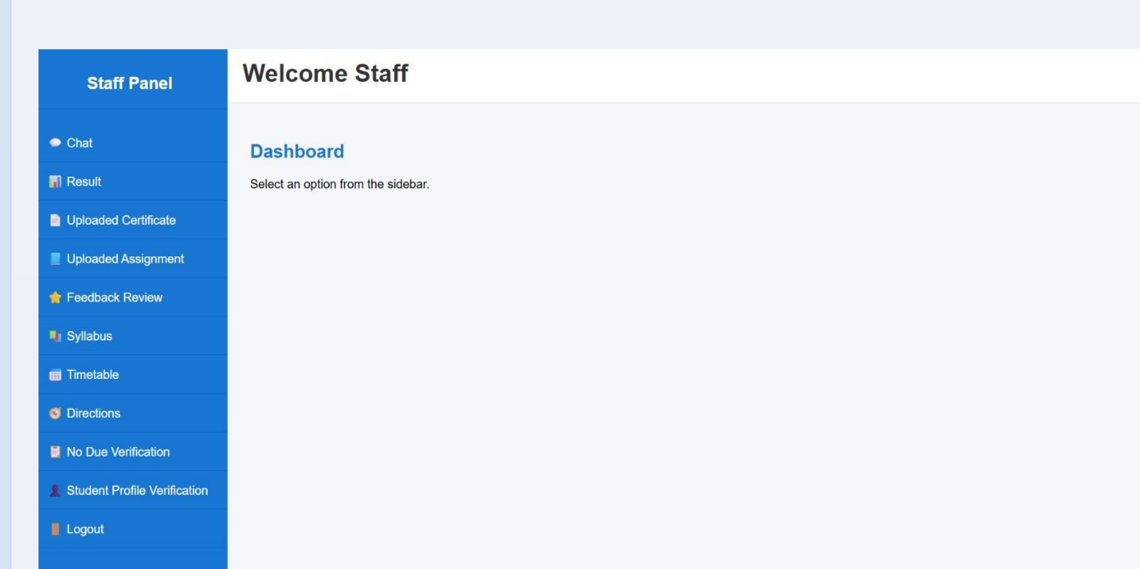
Verification System**:** A student no-due verification module and Feedback System: A feedback module was implemented to collect and manage faculty performance.

9. Data Storage

User credentials, interaction data, and location information are stored in the database for system operation and maintenance.

RESULT





CONCLUSION

This project demonstrates the use of Artificial Intelligence to improve communication within educational institutions. By integrating Natural Language Processing and Machine Learning techniques, the chatbot provides instant and accurate responses to student queries related to academics, administration, and assists users in identifying and locating facilities within a campus environment

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